Recurrent neural networks & transformer networks

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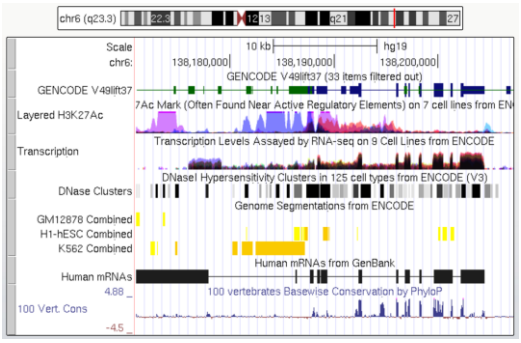
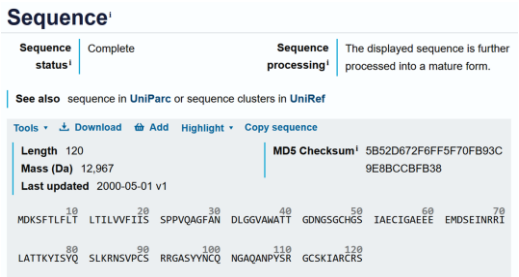
Machine learning methods in biomedical data
Academic year 2025-2026

Sequence data are everywhere in biomedicine

genome data



proteins



genome.ucsc.edu, www.uniprot.org,

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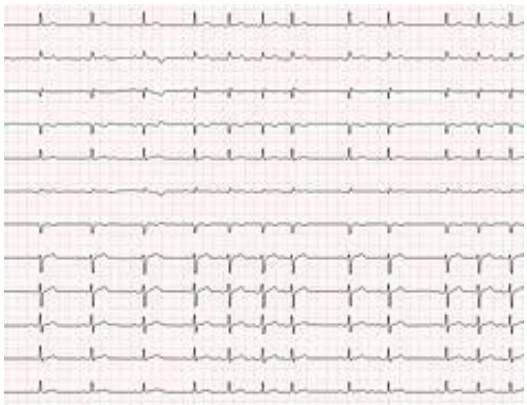
Many biomedical data types are naturally represented as **sequences**. Genomes and proteins are sequences of biological symbols, where the order of nucleotides or amino acids carries functional information. This makes sequence modelling a central problem across modern biomedical research.

Sequence data are everywhere in biomedicine

electrocardiogram



research articles

EXPERIMENTAL PROCEDURES

Training and Evaluation of New MS^{2}PIP Spectrum Prediction Models for Immunopeptides

To train and test new MS^{2}PIP models, five publicly available immunopeptide data sets and one publicly available chymotrypsin-digestion data set were downloaded from PRIDE Archive (29, 30). Similarly, for evaluation on representative unseen data, four distinct data sets were downloaded: (i) a data set containing HLA-I immunopeptides, (ii) a data set containing HLA-II immunopeptides, (iii) the data set with tryptic peptides that was previously used to evaluate the existing MS^{2}PIP higher-energy collision-induced dissociation (HCD) models, and (iv) a data set containing chymotrypsin-digested peptide data. The corresponding ProteomeXchange project identifiers as well as the number of unique peptides and HLA patterns for each data set are listed in supplemental Table S1.

As tandem mass spectrometry (MS2) fragmentation patterns are highly dependent on the instrument, instrument settings, fragmentation method, and any applied labeling methods (25), all MS^{2}PIP train, test, and evaluation data must originate from experiments with the same experimental parameters. Unlabeled HCD data from

Figures: commons.wikimedia.org, 10.1016/j.mcpro.2022.100266

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Not all biomedical sequences are molecular. Signals such as **electrocardiograms** and ordered text such as **research articles** are also sequential data, because meaning depends on what appears earlier or later. Sequence models are therefore useful far beyond genomics and proteomics.

Symbols must first be converted into numbers

Manual feature engineering

ACDTK

	A	C	D	T	K
Basi	37	35	59	129	191
Heli	68	23	33	29	32
Hydr	51	75	25	35	0
pl	32	23	0	27	69

one-hot encoding

ACDTK

	A	C	D	T	K
A?	1	0	0	0	0
C?	0	1	0	0	0
D?	0	0	1	0	0
…	…	…	…	…	…
T?	0	0	0	1	0
…	…	…	…	…	…
K?	0	0	0	0	1
…	…	…	…	…	…

learned embedding

ACDTK

A	C	D	T	K
0.73	0.14	0.88	0.46	0.21
0.57	0.95	0.32	0.69	0.04
0.81	0.27	0.53	0.10	0.76
0.38	0.62	0.17	0.84	0.49
0.92	0.35	0.58	0.07	0.64
0.11	0.79	0.43	0.96	0.28
0.67	0.24	0.85	0.51	0.13
0.40	0.73	0.19	0.60	0.88

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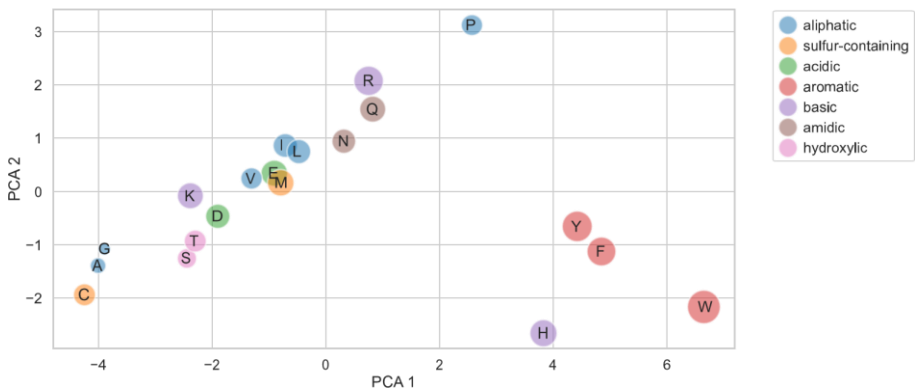
Neural networks cannot work directly on symbolic inputs such as letters or words. These symbols must first be converted into numerical representations, for example through **manual feature engineering**, **one-hot encoding**, or **learned embeddings**. The choice of representation strongly influences what patterns a model can learn.

Learned embeddings can capture biologically meaningful similarity



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A learned **embedding** represents each symbol as a vector of numbers that is optimized during training. Unlike one-hot encoding, embeddings can place biologically similar symbols closer together in representation space. This allows the model to learn that some amino acids, codons, or words behave more similarly than others.

In this example, an embedding has learned to place amino acid close together in its representation space if they have similar physicochemical properties.

In sequence data, order matters

Cage the swallow.

≠

Swallow the cage.

```
MDKSFTLFLTLTILVVFIISSPPVQAGFANDLGGVAWATT
GDNGSGCHGSIAECIGAEEEEMDSEINRRILATTKYISYQ
SLKRNSVPCSRRGASYYNCQNGAQANPYSRGCSKIARCRS
```

≠

```
MDKSFTLFLTLTILVVFIISSPPVQAGFANDLGGVAWATT
GDNGSGCHGSIAECIGAEEDMEESEINRRILATTKYISYQ
SLKRNSVPCSRRGASYYNCQNGAQANPYSRGCSKIARCRS
```

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In sequence data, the **same elements in a different order** can have a different meaning. This is true both for language and for biological sequences, where even a small change in position can alter interpretation or function. Sequence models are designed specifically to preserve and use this ordering information.

In sequence data, memory matters as well

culture
office culture
cell culture

bank
river bank
investment bank

expression
artistic expression
gene expression

cell
immune cell
prison cell

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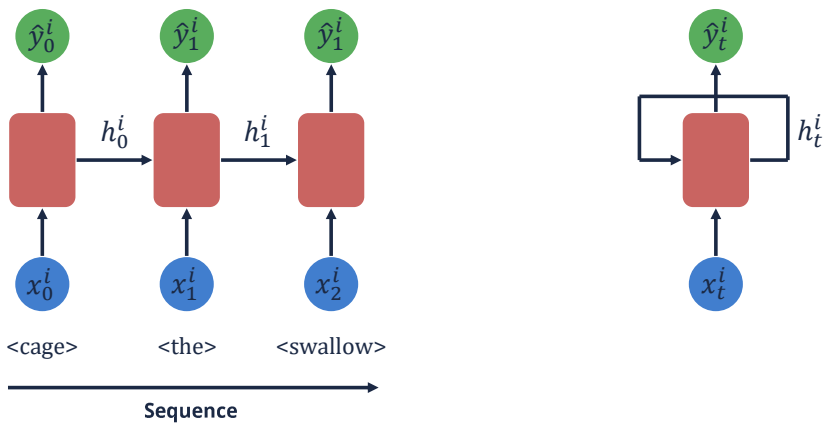
In many sequences, the correct interpretation of one element depends on the **context that came before it**. The same word can have different meanings in different phrases, just as the importance of a residue or motif may depend on the broader biological sequence. A useful sequence model therefore needs not only order, but also some notion of memory.

Recurrent neural networks process a sequence one step at a time



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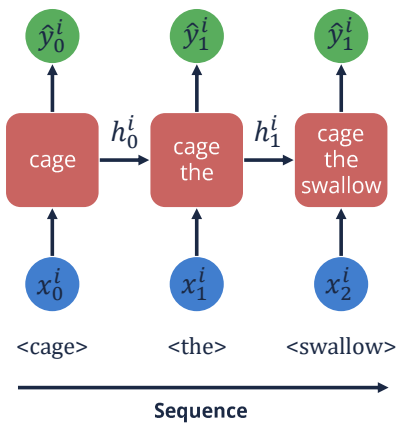
A **recurrent neural network (RNN)** reads a sequence one element at a time and updates its internal state after each step. The same recurrent unit is reused across the sequence, which allows the model to process inputs of variable length. In this way, information from earlier positions can influence later predictions.

The hidden state acts as a running summary of the past



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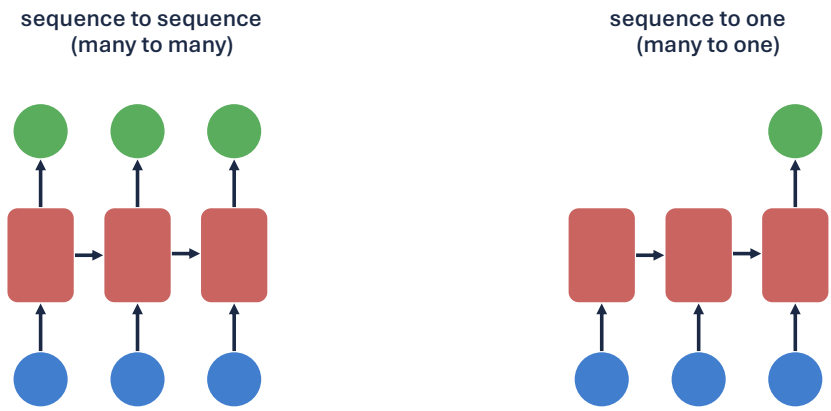
The **hidden state** of an RNN can be viewed as a compact summary of everything the model has seen so far. After each new input, this summary is updated to incorporate the new information. The model therefore carries forward a learned representation of past context while moving through the sequence.

Recurrent neural networks can predict at each step or at the end



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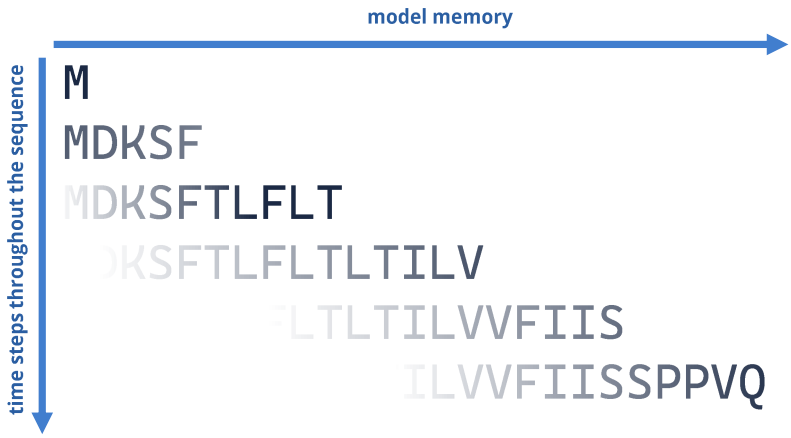
RNNs can be used for different sequence tasks depending on where predictions are made. In **sequence-to-sequence** settings, the model produces an output at each step, while in **sequence-to-one** settings it produces a single output after reading the full sequence. This makes recurrent models flexible for both token-level and sequence-level prediction problems.

Vanilla RNNs struggle with long-range dependencies



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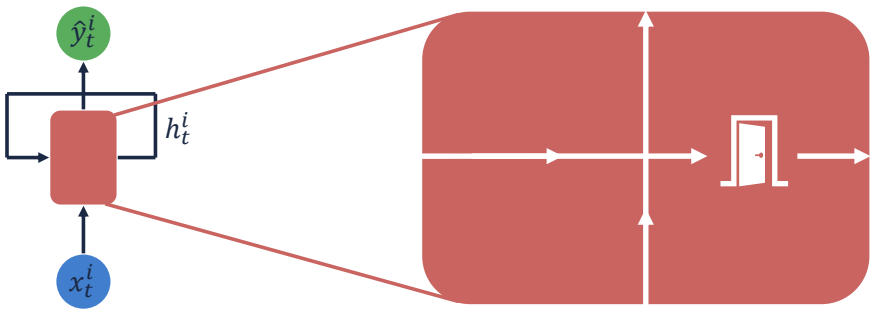
A basic RNN has difficulty preserving information over many time steps. As a sequence becomes longer, relevant earlier information may gradually fade from the model's internal state. This makes it hard for vanilla RNNs to capture **long-range dependencies**, which are common in biological and linguistic data.

LSTMs and GRUs introduced gated memory to better preserve the important information



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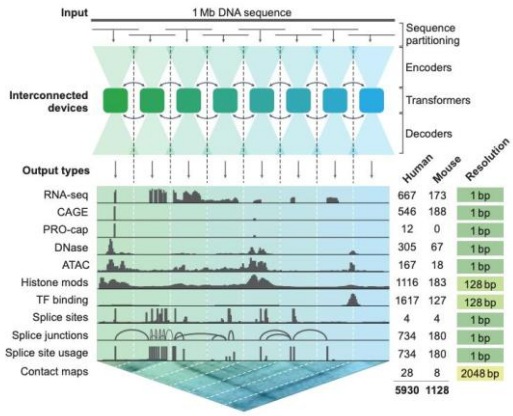
LSTMs and **GRUs** were developed to address the memory limitations of vanilla RNNs. They introduce gating mechanisms that help the model decide what information to keep, update, or forget over time. This makes them more effective when important context must be preserved across longer sequences.

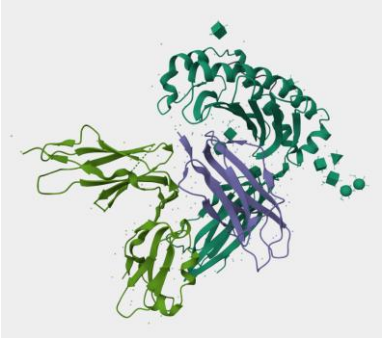
For complex tasks, recurrent memory is often not enough…

Especially when important context is far away

Long-range genome interactions



Amino acid interactions in 3D structure



nature.com/articles/s41586-025-10014-0,uniprot.org/uniprotkb/P60709

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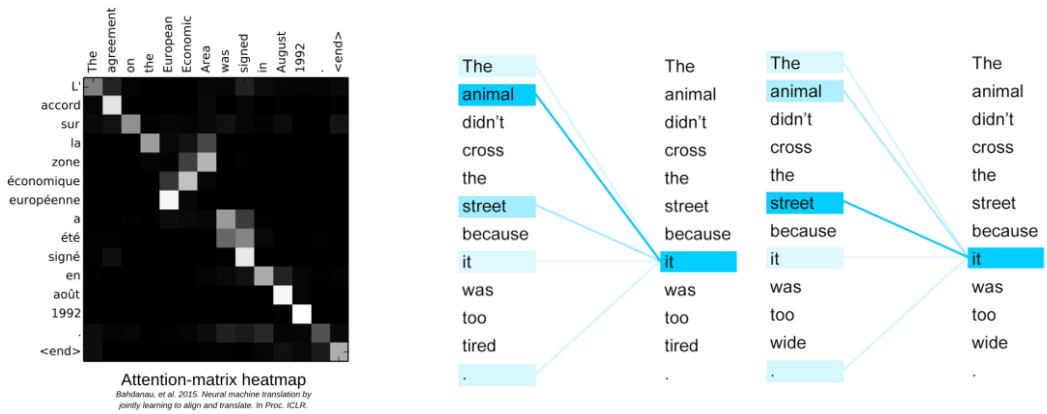
Some sequence problems require the model to relate positions that are **far apart**. In biomedicine, this occurs for example in long-range genome interactions and in proteins, where residues distant in sequence can still interact in three-dimensional space. In such settings, compressing all past information into a single recurrent memory can become a major limitation.

Attention lets the model focus on the most relevant parts of the entire sequence



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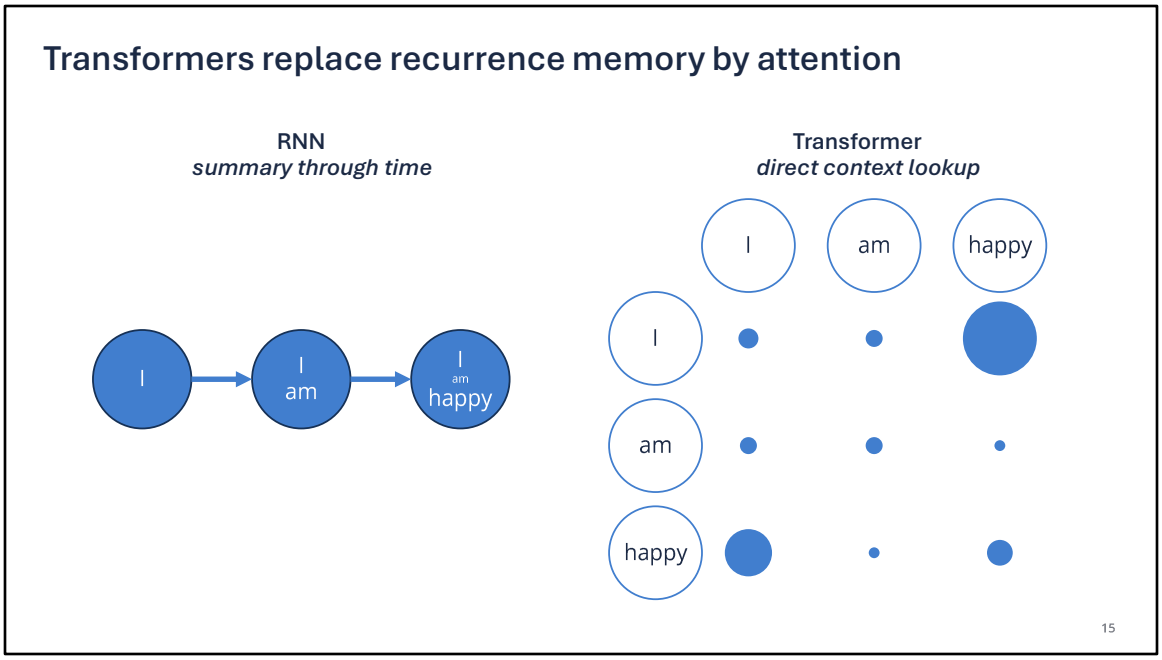
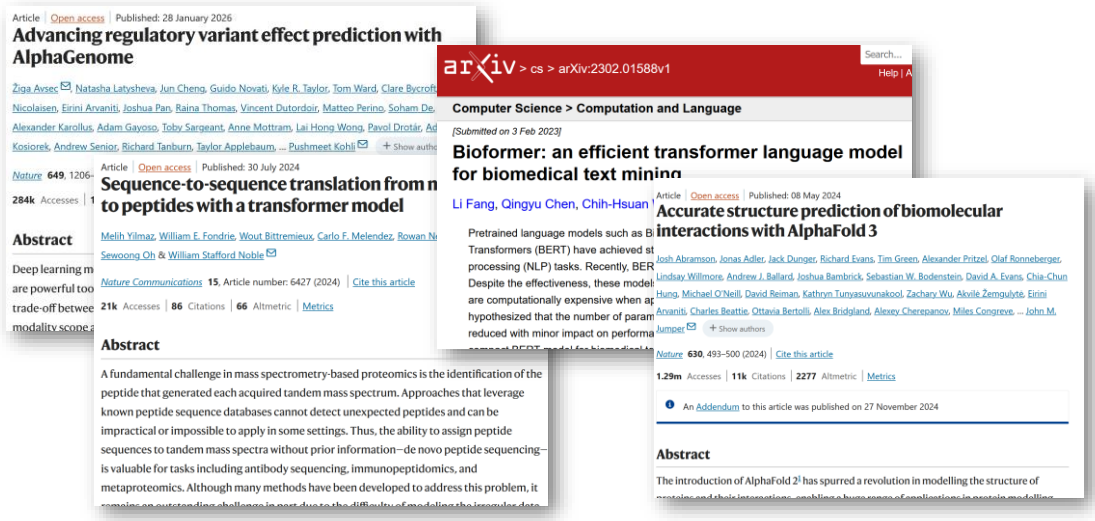
Attention allows the model to assign different importance to different positions in the sequence when making a prediction. Instead of relying only on a compressed memory, the model can look back at the most relevant elements directly. This makes it easier to capture relationships between distant sequence positions.



Transformers move away from recurrent step-by-step memory and instead rely primarily on attention. Rather than summarizing the past into a single evolving hidden state, each position can directly consult other relevant positions in the sequence. This makes transformers particularly powerful for modelling complex dependencies over long distances.

Transformers are now widely used across biomedical data types



Transformer-based models are now used across a wide range of biomedical domains, including **genomics**, **proteomics**, **biomedical language processing**, and related areas. Their strength lies in learning rich sequence representations and in capturing long-range context efficiently. As a result, they have become a major modern architecture for sequence modelling in biomedicine.

Recap

- Sequence data are ordered, so position carries meaning.
- Symbols must be converted into numbers, often through learned embeddings.
- RNNs process sequences step by step and maintain a hidden state as memory.
- Vanilla RNNs struggle with long-range dependencies; LSTMs and GRUs improve this.
- Transformers use attention to relate distant sequence elements directly.